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Earth and Planetary Science Letters 227 (2004) 395–410

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Magnetostratigraphy: the key to a global correlation of the classic Germanic Trias-case study Volpriehausen Formation (Middle Buntsandstein), Central Germany

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Received 10 May 2004; received in revised form 27 July 2004; accepted 4 September 2004
Available online 8 October 2004
Editor: V. Courtillot

Abstract

An integrated lithostratigraphic–magnetostratigraphic study was carried out in order to provide a magnetic record for the Volpriehausen Formation of the classic Germanic Trias. In its type area of Central Germany, the ~150-m thick Volpriehausen Formation consists of nine small-scale fining-upward cycles, ranging from about 10 to 20 m in thickness. They can readily be correlated over large parts of the Central European Basin using a combination of cyclic stratigraphy and gamma ray logging. Based on this robust high-resolution lithostratigraphic framework, a precise positioning and verification of paleomagnetic data has been realized. To this end, the classic outcrop Kleines Bodetal and the continuously cored well Bockenem A100 were collected, yielding a total of 361 paleomagnetic samples. From about 87% of them, a characteristic remanence was obtained in both the magnetite-bearing gray lithologies as well as the hematite-bearing red-brown lithologies. The between-site correlation allows the creation of a well-defined composite magnetostratigraphy of the uppermost Bernburg Formation to lowermost Detfurth Formation, encompassing a total of five magnetic polarity intervals, which can be combined unambiguously with a recently established polarity scale for the uppermost Zechstein and Lower Buntsandstein. Moreover, the resulting composite magnetostratigraphy can be correlated with corresponding records from the Continental, Boreal, and Tethyan realms. Thus, the Buntsandstein magnetostratigraphy is important in that it provides valuable anchors to link the Germanic Trias high-resolution lithostratigraphy to the biostratigraphically calibrated Triassic substages.
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Keywords: magnetostratigraphy; cyclicity; gamma ray logging; Triassic; Buntsandstein; Germany

1. Introduction

The classic Germanic Trias is of international importance in that this geological system was first introduced in the epicontinental Central European

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doi:10.1016/j.epsl.2004.09.011

Basin in terms of lithostratigraphy [1]. Today, the term Triassic is used in the chronostratigraphic sense. In the Tethyan and Boreal realms, its subdivision is based on index fossils, which are rare or absent in the Germanic Trias, especially in the mainly continental Buntsandstein. In terms of biostratigraphy, correlation based on conchostracans (branchiopods) is allowing at least an approximate comparison of the Buntsandstein with the ammonite- and conodont-calibrated marine stratotypes (e.g., [2,3]).

The global synchronous character of magnetic polarity reversals has made magnetostratigraphy an essential tool for precise facies-independent correlation between different depositional environments. In the last two decades, significant progress has been made in constructing a geomagnetic polarity scale for the Lower Triassic (e.g., [4–12]). Hitherto, in the German type, area of the Buntsandstein paleomagnetic research has been strongly neglected. Without exception, all previous studies focused on its coarse-grained marginal facies of South Germany [13–16], which is less suitable as paleomagnetic recorder, because of widespread remagnetization. Besides, it is characterized by significant time gaps due to discontinuous sedimentation and erosion, respectively.

Consequently, recent investigations were restricted to the more complete fine-grained basinal facies of the Buntsandstein [17–19], applying an integrated lithostratigraphic–magnetostratigraphic method. By using a combination of cyclic stratigraphy and gamma ray logs (GR logs), a high-resolution lithostratigraphy of the Buntsandstein has been established (e.g., [20–22]). Based on this robust lithostratigraphic framework, a precise positioning and verification of paleomagnetic data can be achieved, resulting in a well-defined composite magnetic record. Recently, this comprehensive method was successfully applied to the uppermost Zechstein and Lower Buntsandstein (e.g., [17,18]).

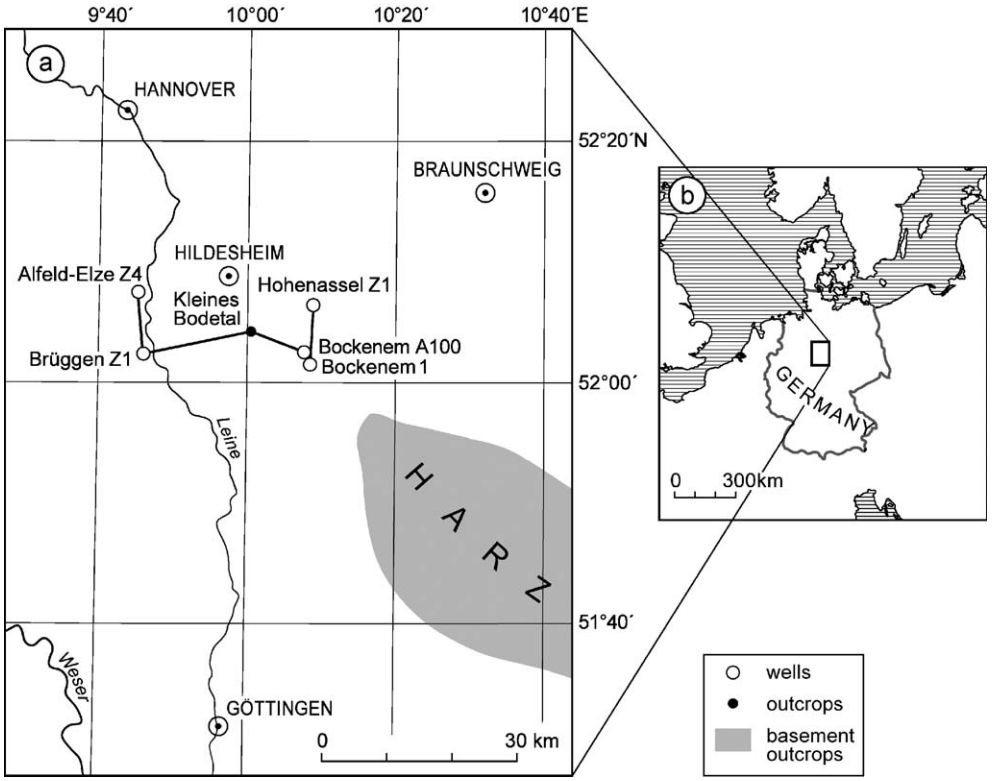


Fig. 1. (a) Simplified map of Central Germany with the location of the outcrop and wells investigated. Location of wells used for gamma ray log cross-section (see also Fig. 4). (b) Position of map shown in panel (a).

The aim of this study is to establish a geomagnetic polarity scale for the Volpriehausen Formation in its basinal type area of Central Germany in order to extend the Lower Buntsandstein magnetostratigraphy [18] into the Middle Buntsandstein. To this end, the classic outcrop Kleines Bodetal (e.g., [23]) and the continuously cored well Bockenem A100 were investigated using both lithostratigraphic and magnetostratigraphic tools (Fig. 1).

Along with biostratigraphic data, the magnetic record of the Buntsandstein is important in that it eventually provides a means for precise correlation of biostratigraphically calibrated Triassic substages with the high-resolution lithostratigraphy of the classic Germanic Trias. Besides, it considerably contributes to an improved dating of sequence stratigraphic units (e.g., [17,24,25]) as well as understanding of possible interbasinal controlling factors on sedimentation, for instance, Milankovitch orbital forcing (e.g., [26,27]).

2. Geological setting

The Buntsandstein represents the lower group of the tripartite classic Germanic Trias [1]. In terms of lithostratigraphy, it is traditionally subdivided into three subgroups [23]: (a) the fine-grained Lower Buntsandstein comprising characteristic oolites, (b) the Middle Buntsandstein composed of fine- to coarse-grained sandstones and shales, and (c) the pelitic Upper Buntsandstein containing evaporites (Fig. 2).

In Central Germany, the Volpriehausen Formation of the fourfold Middle Buntsandstein is 140- to 200-m thick, mainly consisting of brown sandstones and shales, which were deposited during the Early Triassic in predominantly fluvio-lacustrine in marine-influenced environments of the large epicontinental Central European Basin (CEB). Traditionally, the Volpriehausen Formation is further subdivided into the

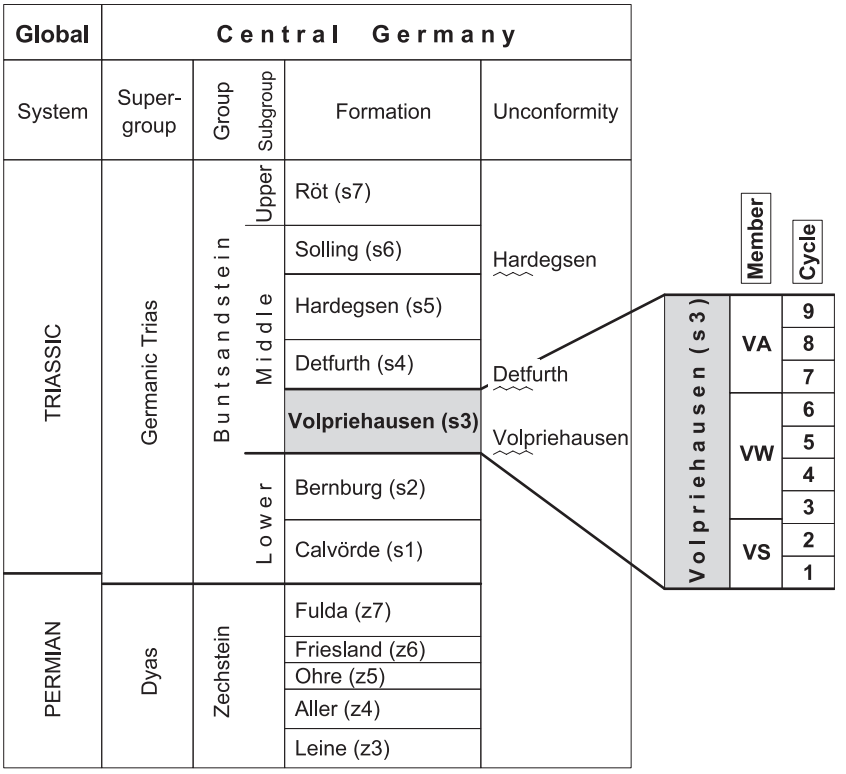


Fig. 2. Lithostratigraphy of the uppermost Zechstein and Buntsandstein in Central Germany with the Volpriehausen Formation, consisting of nine small-scale fining-upward cycles (this study); VS="Volpriehausen–Sandstein" Member, VW="Volpriehausen–Wechselfolge" Member, VA="Volpriehausen–Aviculaschichten" Member.

"Volpriehausen–Sandstein" (VS), the "Volpriehausen–Wechselfolge" (VW), and the "Volpriehausen–Aviculaschichten" (VA) Members (Fig. 2; [23]). The lower boundary of the Volpriehausen Formation is located at the base of the first basin-wide influx of coarse-grained sands, forming the VS Member. Its top is marked by the regional low-relief angular Detfurth unconformity, overlain by the relatively coarse-grained "Detfurth–Sandstein" (DS) Member of the succeeding Detfurth Formation.

The clastic succession shows a distinct cyclicity of varying magnitude, pragmatically subdivided into approximately 10- to 20-m thick small-scale fining-upward cycles. Commonly, they begin with medium- to coarse-grained sandstones that grade into more pelitic sediments. The cycles can be used for correlation in cores, outcrops, and wireline logs. With the latter (e.g., gamma ray logs), they can be traced over large parts of the CEB (e.g., [21]).

Additionally, the correlation is supported by several prominent marker beds, for example, the "Görtelsandstein" ("Görtel Sandstone"; [28]).

Alternatively, the small-scale cycles of the Buntsandstein can be interpreted as more or less symmetrical base-level cycles [17], which are considered to reflect climatic fluctuation of alternating drier and wetter periods due to solar-induced ~100-ka Milankovitch eccentricity cycles (e.g., [21,27]). In terms of sequence stratigraphy, the Volpriehausen Formation represents the second third-order sequence of the Buntsandstein [24].

The biostratigraphic subdivision of the Buntsandstein is mainly based on conchostracans, with the Volpriehausen Formation belonging to a lower *Magniestheria quellaensis–Lioleaiina radzinskii* zone, encompassing the VS Member and an upper *Liograpta* (M.) *mangaliensis* zone, comprising the VW and VA Members (e.g., [2]).

3. High-resolution lithostratigraphy

3.1. Method of gamma ray logging

The natural gamma radiation of the Kleines Bodetal outcrop was recorded using a portable scintillation counter (G.B.-H. Elektronik). For this, the section was measured statically every 10 cm over a selected time integration of 10 s, resulting in a higher resolution than the dynamically measured wireline gamma ray logs (GR logs). The uncalibrated outcrop measurements are stored in the unit as counts per second (cps). In general, the GR logs of the Buntsandstein sections basically display the content of ^{40}K (e.g., [29]), which above all is present in clay minerals. Therefore, the logs reflect more or less the clay content ("shaliness") of the clastic succession, which is displayed by higher radioactivities. Accordingly, with the help of GR logging, it is possible to calibrate the log to lithology and to directly compare outcrop geology with wireline logs from wells (e.g., [17,22]).

3.2. Results of GR logging and cyclic stratigraphy

The combination of GR logging and cyclic stratigraphy results in a precise correlation of outcrops with cored and uncored wells, respectively. This integrative method is exemplified for the uppermost Volpriehausen Formation and the lowermost Detfurth Formation in both the Kleines Bodetal outcrop and the Bockenem A100 core (Fig. 3). The selected interval comprises the upper half of cycle 8 (subcycles c and d) and cycle 9 (subcycles a to c) of the VA Member as well as the lowermost part of the DS Member. All cycles and subcycles, respectively, begin with medium- to coarse-grained sandstones that grade into siltstones and claystones. They are readily obvious in the GR logs, with the sandy lower part being indicated by very low radiation (35 to 65 cps), and the shaly upper part being displayed by distinctly higher values (70 to 100 cps). In general, the sandstones are characterized by 40 to 60 cps, whereas the pelitic sediments vary between 80 and 100 cps. Highest values (>90 cps) are restricted to green-gray claystones, for example, the significant double peak in the lower part of subcycle 8d, which belongs to the green-gray interval 4 (Fig. 3). Commonly, the coarse-grained sandstones have even less than 40 cps, serving as good litho and gamma markers, for example, the DS Member. Its base, which is indicated by a significant decrease in radiation from more than 70 to less than 35 cps, is defining the boundary between the Volpriehausen and Detfurth Formations.

By integration of cyclic stratigraphy and GR logging, both sections can readily be compared

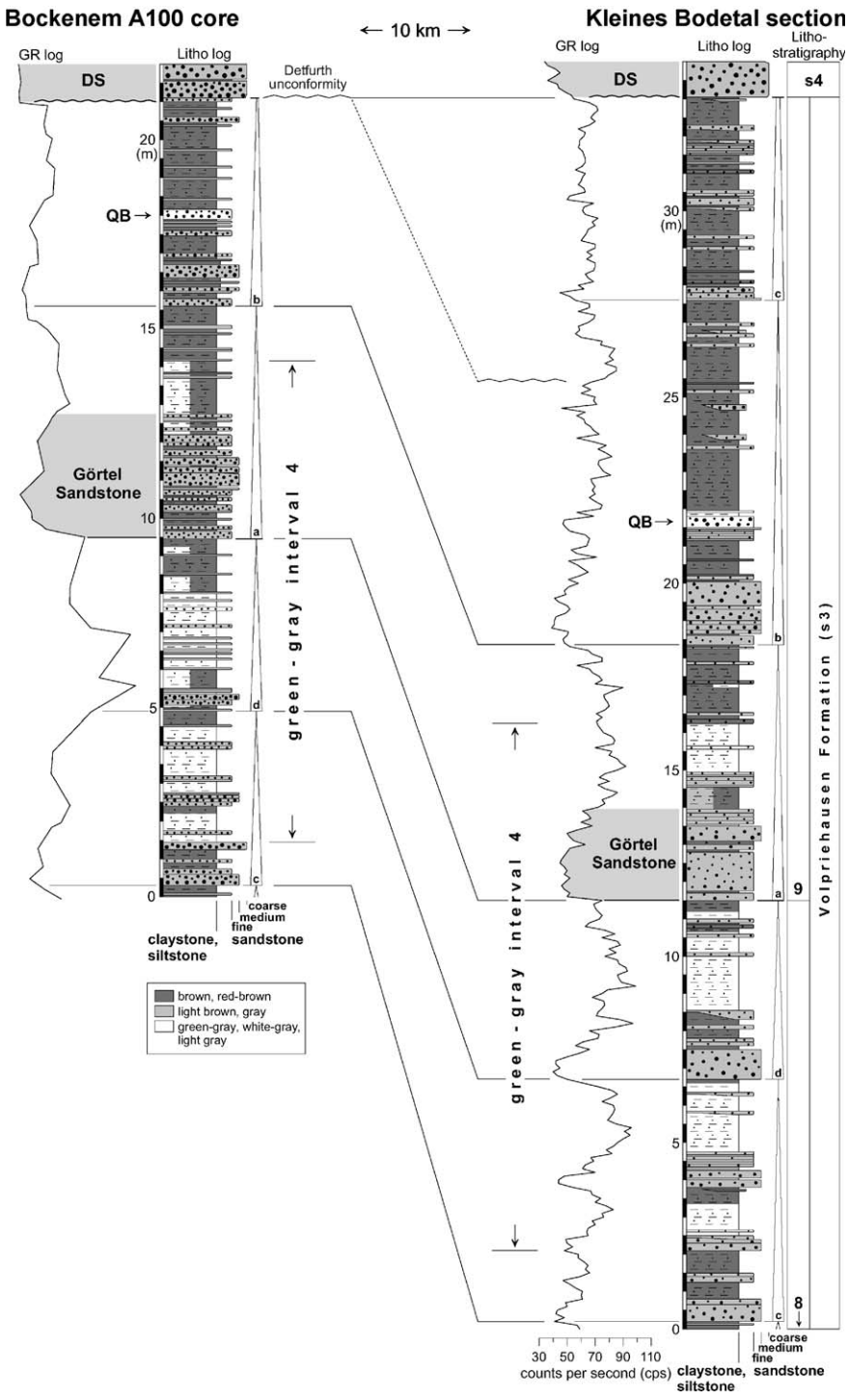


Fig. 3. High-resolution cyclic stratigraphy and GR log correlation between Bockenem A100 core and Kleines Bodetal section; cycles are indicated by numbers, and subcycles are indicated by letters. QB="blendendweiße Quarzitbank" ("shining white quartzitic sandstone"), DS="Detfurth-Sandstein" Member. See Fig. 2 for key.

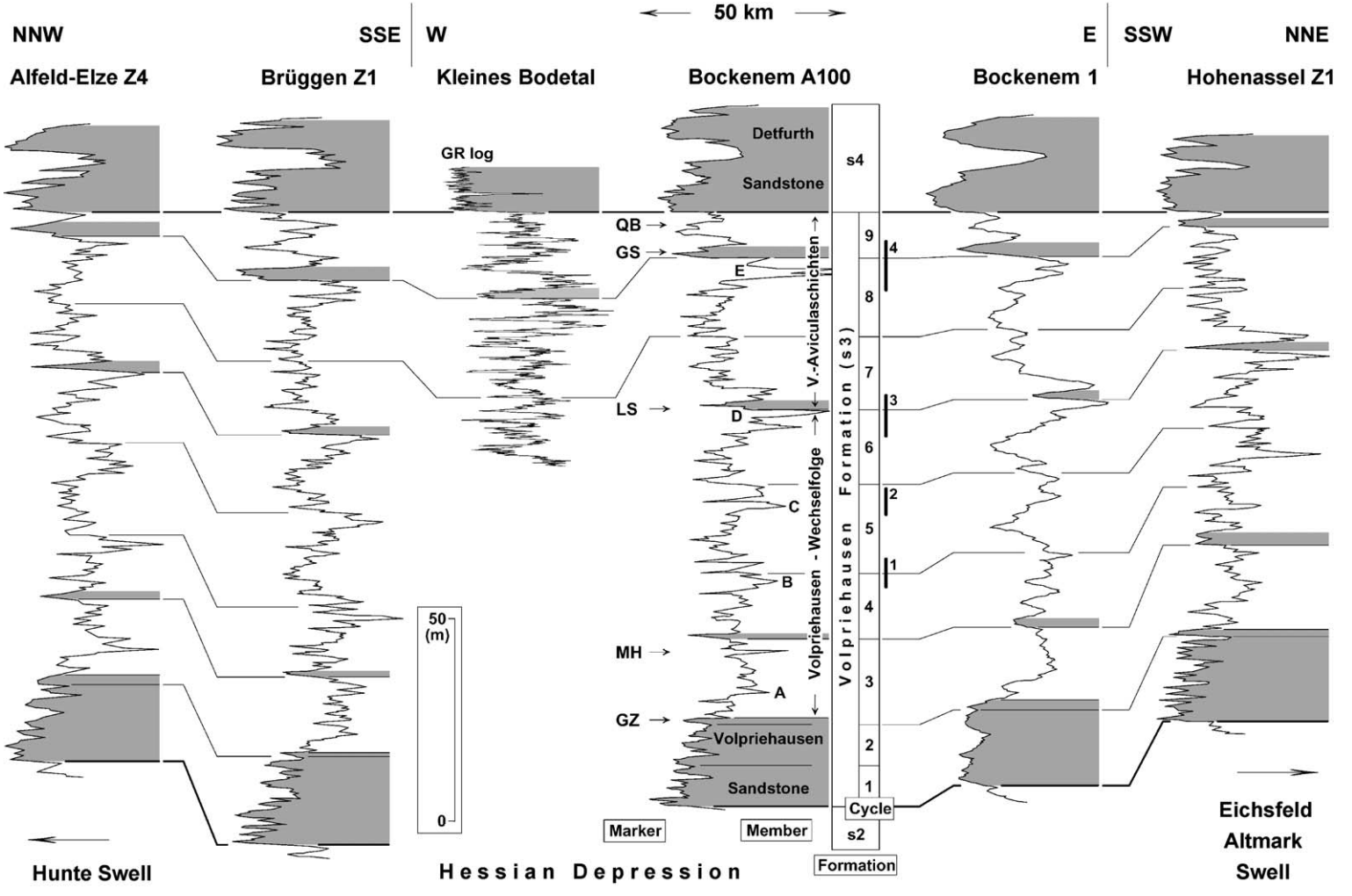


Fig. 4. GR log correlation of the Volpriehausen Formation in an approximately W–E trending transect crossing the paleogeographical elements Hunte Swell, Hessian Depression, and Eichsfeld–Altmark Swell (see also Fig. 1). English terms for displayed marker beds are informal. QB="shining white quarzitic sandstone," GS="Görtelsandstein" ("Görtel Sandstone"), LS="Liograptasandstein" ("Liograpta Sandstone"), MH="Malachit-Horizont" (="Malachite Horizon"), GZ="Grenzsandstein" ("Boundary Sandstone"); 1–4=green-gray intervals 1 to 4; A–E=high-radiation peaks. See also Fig. 2.

(Fig. 3). Besides, this detailed correlation makes possible to calculate the gap below the Detfurth Formation. Accordingly, in the Bockenem A100, core are missing some 7.5 m, comprising uppermost subcycle 9b as well as subcycle 9c of the Volpriehausen Formation.

Moreover, the GR logs allow to precisely correlate the small-scale fining-upward cycles over large parts of the CEB (e.g., [17,20–22]), providing a robust high-resolution lithostratigraphic framework. Based on regional log correlations, there is no evidence for considerable hiatuses within the basinal facies of the Volpriehausen Formation (e.g., [21]), for instance, demonstrated by a 50-km long log transect, crossing the paleogeographical elements of the Hunte Swell, Hessian Depression, and the Eichsfeld–Altmark Swell (Fig. 4). A considerably refined subdivision of the traditionally threefold Volpriehausen Formation now reveals nine small-scale cycles, partly differing from previous proposals, which are based exclusively on wireline logs [20,21]. Whereas the mainly coarse-grained VS Member, comprising cycles 1 and 2 as well as the base of cycle 3, is emphasized by noticeable low radiation, the overlying VW Member (cycles 3 to 6) is indicated by higher radioactivities, which are obvious by the so-called markers A to C [30]. The succeeding VA Member (cycles 7 to 9) is characterized by the mass occurrence of the bivalve *Avicula murchisoni*. Here, the base of the VA Member is placed at the base of cycle 7, being emphasized by a double peak of remarkable high radiation, the marker D, which is readily visible in wireline logs as well as in outcrops. This new boundary is coincident with the "maximum flooding surface" of the second third-order cycle of the Buntsandstein [24], which comprises the complete Volpriehausen Formation. In the Bockenem section, it is located some 13 m below the original base of the VA Member [23], which, in general, is imprecisely defined either at the base of an interval rich in *A. murchisoni* or by different lithological criteria.

Thus, with an integrative litho- and log-stratigraphic study, a precise positioning of paleomagnetic data from the Buntsandstein has been realized (e.g., [17–19,22]), forming the imperative base for a composite geomagnetic polarity scale.

4. Magnetic properties

4.1. Sampling and laboratory procedures

Samples were collected with a portable water-cooled rock drill. In general, fine-grained sandstones, siltstones, and claystones, judged to be the most favorable lithologies, were recovered at 1- to 3-m intervals, yielding a total of 361 oriented-standard (~11 cm^3) paleomagnetic samples. Whereas specimens from red-brown lithologies were subjected to thermal demagnetization, samples from gray lithologies were predominantly demagnetized by alternating fields. The thermal treatment was performed using an ASC Scientific TD48 oven. Measurements of natural remanent magnetization (NRM) were made with either an AGICO JR5A spinner magnetometer (noise level=2×10^{-6} A/m) or a 2G Enterprises automatic DC-SQUID 755SRM cryogenic magnetometer (noise level=1×10^{-6} A/m). The magnetic susceptibility was routinely recorded after each demagnetization step with a Bartington MS2B sensor in order to monitor mineralogical alteration of the magnetic fraction due to heating. Alternating field demagnetization of the gray lithologies was performed using the in-line triaxial alternating field demagnetizer of the cryogenic magnetometer. The direction of the characteristic remanent magnetization (ChRM) was determined using principle component analysis (PCA, [31]), whereas the mean direction of the Kleines Bodetal site was calculated by standard Fisher statistics [32].

Isothermal remanent magnetizations (IRM) were applied with a 2G Enterprises 660 pulse magnetizer and measured with a Molyneux MiniSpin fluxgate magnetometer (noise level=0.2×10^{-3} A/m). Samples were exposed stepwise to peak fields of 10 to 2700 mT along the positive z axis of the specimens in order to record complete IRM acquisition curves. For a more conclusive interpretation, IRM experiments were combined with subsequent thermal demagnetization of IRM inasmuch as magnetic minerals exhibiting similar coercivities generally are characterized by different unblocking temperatures.

4.2. Rock magnetic properties

A representative suite of about 10% of all samples was selected for IRM experiments. The corresponding

acquisition curves establish significant differences in the ferromagnetic mineralogy between the gray and red-brown lithologies, similar to results recently obtained from the Lower Buntsandstein [18]. Usually, the gray samples contain a low-coercivity fraction, acquiring the majority of IRM between 100 and 200 mT and being unblocked close to 590 °C, which can be ascribed to magnetite (Fig. 5a). Commonly, most of IRM is reached in ambient fields less than 500 mT.

In general, the red-brown lithologies are dominated by the presence of a high-coercivity component (Fig. 5b). It is saturated between 600 and 1000 mT, having an accompanying maximum unblocking temperature of 675 °C, which is attributed to hematite. Additionally, in some red-brown samples, an inflexion in the acquisition curves is displayed at ~300 mT, indicating the presence of an additional low-coercivity fraction (e.g., Fig. 5c). It acquires most of the IRM in fields less than 200 mT, exhibiting a noticeable drop in the demagnetization curve between 500 and 600 °C, identified as most probably magnetite (Fig. 5c). Furthermore, most of the red-brown lithologies are

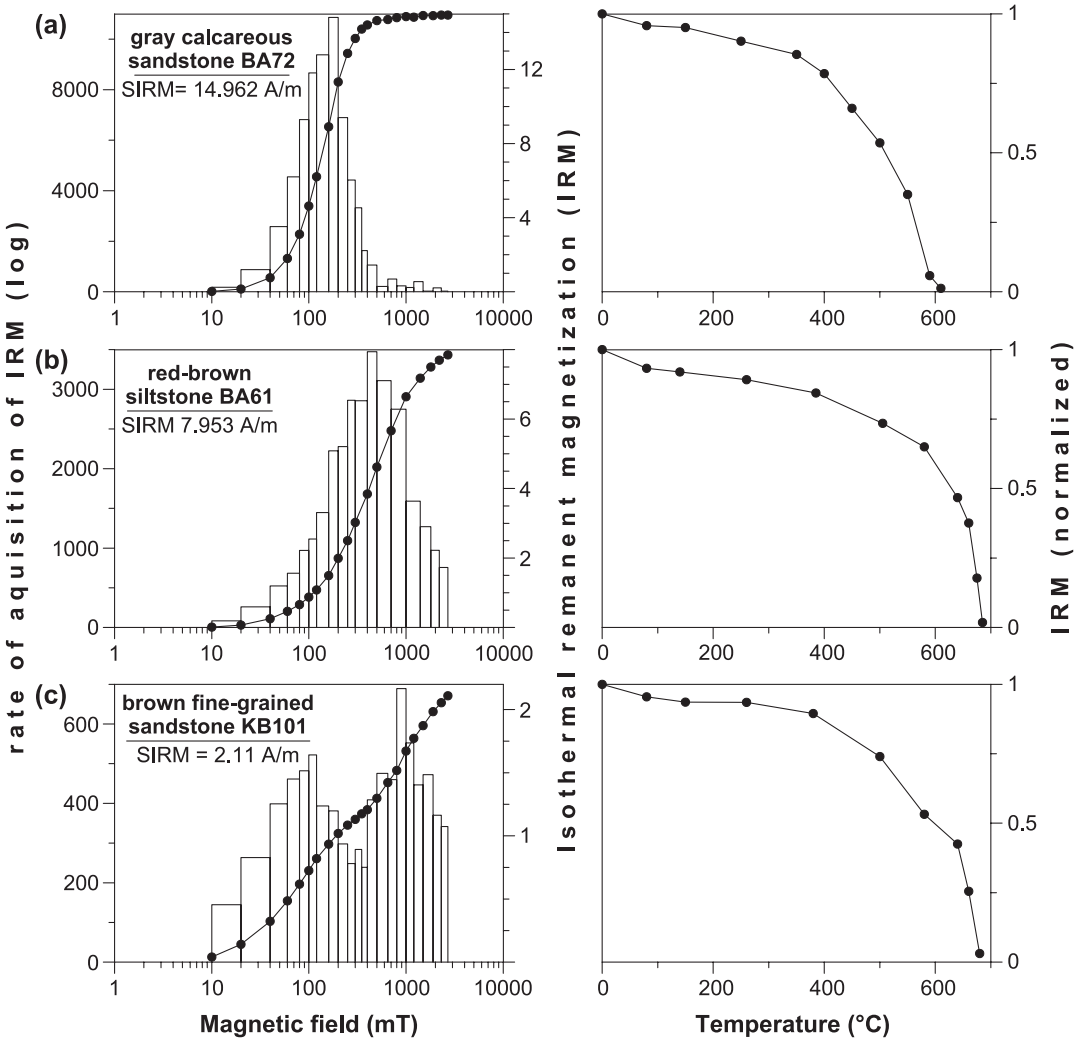


Fig. 5. (a–c) IRM acquisition in fields up to 2700 mT and subsequent stepwise thermal demagnetization of IRM for three representative lithofacies types from the Volpriehausen Formation. Samples are identified with abbreviation of section name (KB=Kleines Bodetal, BA=Bockenem A100) and chronological sampling number.

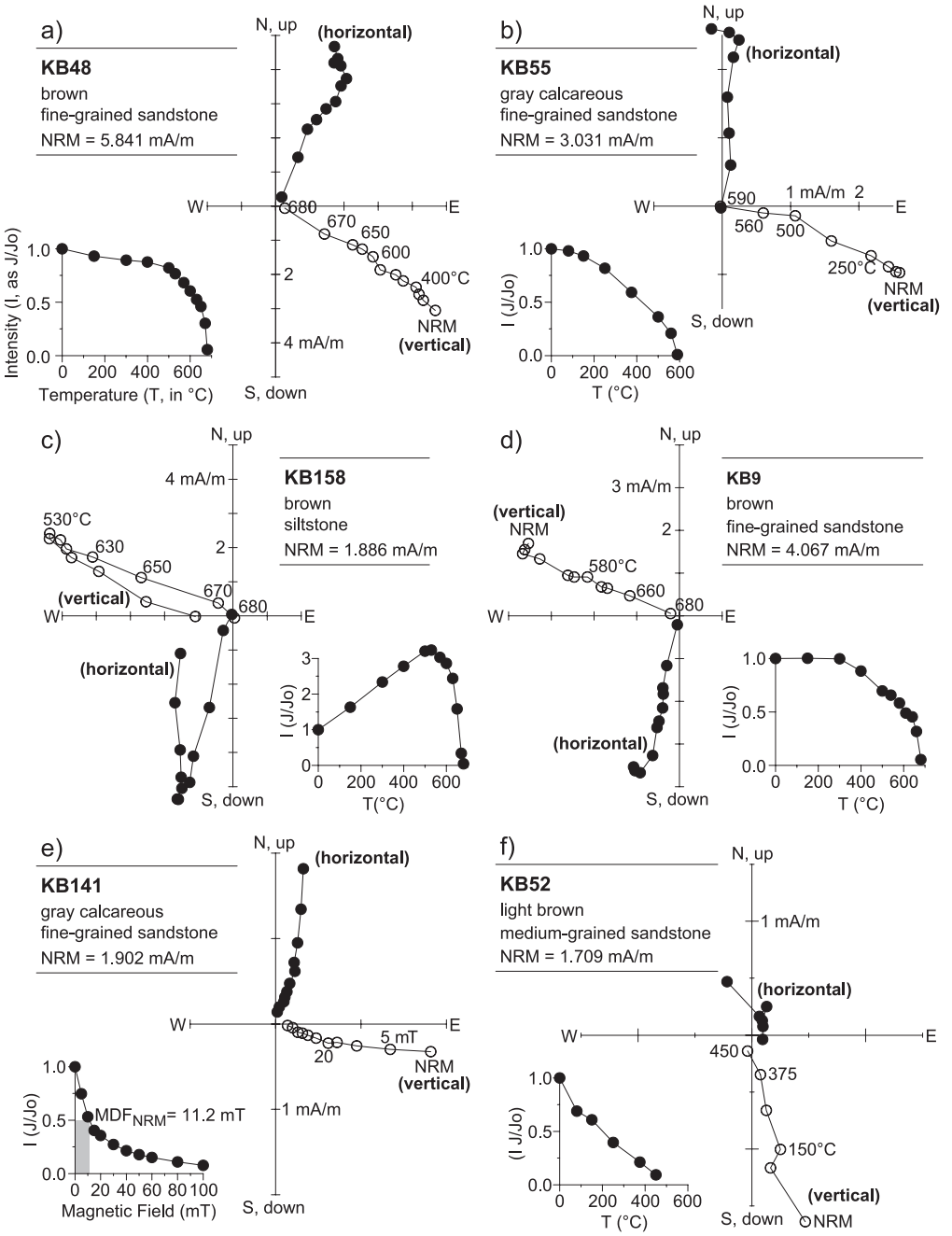


Fig. 6. (a–d) Vector endpoint diagrams and normalized intensity decay curves illustrating thermal and alternating field demagnetization of NRM for six representative samples. Demagnetization results assigned to type I (a–e) and type II (f) magnetizations after tectonic correction. In general, demagnetization of type I magnetizations (a–d) at first removes a steeply inclined and northerly directed present-day field overprint (component A) at temperatures below 500 °C. Component B, usually converging towards the origin, is isolated over a discrete unblocking temperature range from about 500 to 590 °C or an alternating field of 100 mT, indicating magnetite (b), and from about 580 to 680 °C, indicating hematite (a, c, and d), respectively. Direction of component B is regarded as primary magnetization (ChRM) being close to the expected Early Triassic geomagnetic field direction for Central Germany. The results of type II magnetizations (f) reveal only a steep present-day field overprint. See Fig. 5 for key.

not entirely saturated at maximum field of 2700 mT (e.g., Fig. 5b–c). Such specimens often show a significant drop in the demagnetization curve between 50 and 100 °C (e.g., Fig. 5b), indicating the high-coercivity mineral goethite.

4.3. Paleomagnetic properties

In all, 347 specimens were demagnetized thermally or by alternating fields. The NRM intensities range between 0.1 and 15 mA/m, with the red-brown lithologies, in general, displaying higher intensities than the gray lithologies. Values of initial susceptibility vary between 25 and 310×10^{-6} (SI).

Usually, the NRM of the red-brown lithologies is indicated by two superimposed components of magnetization, related to lower and higher unblocking temperature ranges (Fig. 6a, c–d). Component A is characterized by northerly declinations and steep positive inclinations (60°–70°), ascribed to a viscous overprint due to the local present-day field direction (D=0°, I~66°), which is commonly removed below 500 °C. The remaining component B, indicated by a maximum unblocking temperature of 680 °C, yielded two antipodal groups of directions: (a) northeasterly declinations (Fig. 6a) with shallow positive inclinations (10° to 40°) and (b) southwesterly declinations (Fig. 6c–d) with shallow negative inclinations (−10° to −40°). In the vector endpoint diagrams, they are revealed as linear trajectories towards the origin. Component B is regarded as a ChRM, usually being isolated over an unblocking temperature range from 500 to 680 °C but most commonly above 580 °C, indicating hematite as carrier of remanence. The mean paleomagnetic direction of component B from the Kleines Bodetal site (D=14.4°, I=24.7°, α95=3.8°) is close to the expected Early Triassic geomagnetic field direction for Central Germany [33,34] and similar to results recently obtained from the Lower Buntsandstein [17,18].

Normally, the NRM of the gray lithologies also consists of two components (Fig. 6b, e). First, removed with either a maximum unblocking temperature below 500 °C or an alternating field of 20 mT is a northerly directed relatively steep inclination, which is identified as a recent overprint and referred to as component A. The remaining magnetization (component B), which is regarded as ChRM, reveals both polarities: northeasterly declinations with shallow positive inclinations and southwesterly declinations with shallow negative inclinations. Component B is isolated over an unblocking temperature range from 500 to 590 °C, indicating magnetite.

In all, 301 samples reveal a primary direction comparable to results recently obtained from the Lower Buntsandstein and accordingly named type I magnetizations [18]. They were used to construct a well-documented composite magnetic record for the Volpriehausen Formation. In all, 46 samples exhibit, at best, only a single remanence direction, converging towards the origin. It is readily removed by alternating fields of up to 20 mT and by thermal demagnetization below 500 °C, revealing steep positive inclinations (50° to 70°), which are displaying the local present-day field direction for Central Germany (Fig. 6f). Demagnetization to higher temperatures yielded often only scattered directions. Therefore, no identifiable characteristic remanence of Early Triassic origin is apparently present in these samples (type II magnetizations). In almost all cases, the more permeable medium-grained sandstones, collected only to check magnetic properties, revealed exclusively such type II magnetizations.

5. Magnetostratigraphy and discussion

Previous data indicate that an Early Triassic remanence is preserved in type I magnetizations. They are associated with components that are isolated at high unblocking temperatures, indicating magnetite in most of the gray and hematite in the red-brown lithologies, respectively. These ChRMs are used to establish a reasonable well-defined magnetostratigraphy for the Volpriehausen Formation. About 63 m of the upper Volpriehausen Formation and some 12 m of the overlying Detfurth Formation were sampled in the Kleines Bodetal section, yielding a total of 145 specimens. In all, about 88% of them reveal a dual polarity magnetization (ChRM) close to the expected direction for the Early Triassic (Fig. 7). From the continuously cored Bockenem A100 well, spanning the uppermost 11 m of the Bernburg Formation, the complete Volpriehausen Formation (146 m) and about 35 m of the overlying Detfurth Formation were recovered 216 samples, with 86% of them exhibiting a ChRM.

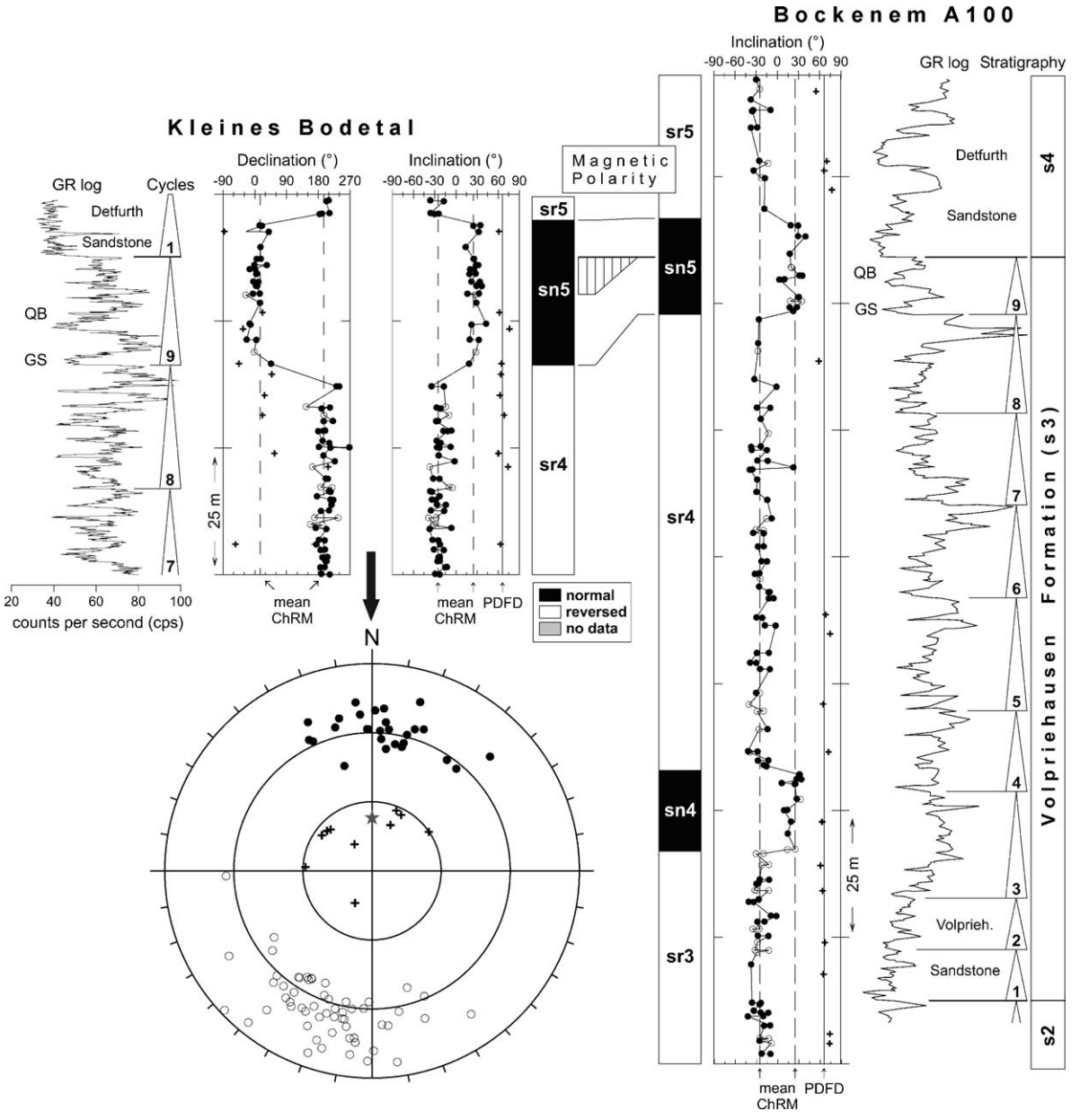


Fig. 7. Between-site correlation of magnetic records of the Kleines Bodetal and Bockenem A100 sections, with declinations and inclinations versus depth. Type I magnetizations are used to interpret polarity in which ChRM carried by hematite is characterized by solid symbols and those carried by magnetite by open symbols, respectively. Indicated by crosses are type II magnetizations, which reveal steep positive inclinations close to the present-day field direction (PDFD). Equal-area projection after bedding tilt correction of characteristic component directions from Kleines Bodetal section. ○ (●) denotes upward (downward) pointing inclinations of type I magnetizations. Type II magnetizations are indicated by crosses and local PDFD by an asterisk, respectively. For identification of magnetozones, the nomenclature established for the Lower Buntsandstein is carried on [18]. Correspondingly, each constituent polarity zone is labelled by the acronym s for the Buntsandstein and has a suffix for the polarity (n is normal, r is reversed), which is extended by a number,allocated in ascending numerical order from the base of the Buntsandstein. See Figs. 2 and 4 for key.

From the studied interval, five well-documented magnetic polarity zones are delineated (Fig. 7). The uppermost Lower Buntsandstein from the Bockenem site belongs to a reversed magnetozone assigned as sr3. It extends into the lowermost Volpriehausen Formation, comprising cycles 1 and 2 as well as the lower part of cycle 3. The lowermost 6 m of the coarse-grained VS Member were either not collected or show type II magnetizations. Recently, this interval was studied in detail in the nearby Marienburg section, revealing exclusively reversed polarity [17,18]. The upper part of cycle 3 as well as the lower third of cycle 4 belong to thin magnetozone sn4, which is succeeded by the thick reversed interval sr4. Strikingly, the latter extends to the boundary of cycles 8 and 9, spanning some 90 m. In addition, sr4 is also delineated in the Kleines Bodetal section, exhibiting the same lithostratigraphic range. The following magnetozone sn5 encompasses cycle 9 of the Volpriehausen Formation as well as the lowermost Detfurth Formation. Within the lowermost DS Member is located the base of the succeeding reversed polarity zone sr5. In terms of magnetostratigraphy, the gap between the Volpriehausen and Detfurth formations is more extended at the Bockenem site, in that sn5 comprises some 7.5 m more of the uppermost Volpriehausen Formation at the Kleines Bodetal site. This impressively confirms the results obtained from cyclic stratigraphy and GR log correlation (Figs. 3 and 4).

Combined with the recently established magnetostratigraphy for the uppermost Zechstein and Lower Buntsandstein [17,18], the composite magnetic polarity sequence from Central Germany now comprises almost the lower half of the Buntsandstein (Figs. 2 and 8), with both contributing records overlapping some 40 m. A total of 13 polarity intervals, ranging from about 15 to 125 m in thickness, have been delineated from the intersection correlation. The thick normal magnetozone sn1, which is straddling the base of the Germanic Trias, correlates with polarity zone Tbn1 from the Polish part of the CEB [9]. It is succeeded by an interval of predominantly reversed polarity in both the German (sr1 to sr5) and the Polish (Tbr1 to Tbr4) records. Inasmuch as the Lower Buntsandstein corresponds to most of the Polish Baltic Formation, except its lowermost part (e.g., [35,36]), the normal polarity intervals sn1 to sn3 are most probably equivalents of Tbn1 to Tbn3 [18]. The base of the Middle Buntsandstein is located within sr3 and Tbr3, respectively. In terms of lithostratigraphy, the Volpriehausen Formation at least corresponds to the lower part of the Pomorska Formation (e.g., [36]), being consistent with the magnetic polarity patterns (Fig. 8). Consequently, sn5 most probably correlates with Tbn4 from the Pomorska Formation. Apparently, polarity zone sn4 from the lower Volpriehausen Formation has no equivalent in the Polish record, but the corresponding interval of the Pomorska Formation is not completely investigated, for instance, revealing a data gap of some 30 m in the Ko^{3}obrzeg IG-1 and Otyn IG-1 wells [9]. Summarily, the Lower Buntsandstein in Central Europe is characterized by predominantly normal polarity, while the succeeding interval encompassing the Volpriehausen Formation and lowermost Detfurth Formation is mainly of reversed polarity (e.g., [9,15]).

The remarkable thick normal magnetozone sn1 represents a first-order feature, which is visible in virtually all magnetic records across the PTB (e.g., [4–6,8,10,11,18,37–39]). Therefore, it is a valuable anchor to faciliate the comparison of Boreal and Tethyan magnetic records as well as to link the Buntsandstein lithostratigraphy to the biostratigraphically calibrated Lower Triassic substages [18]. With regard to the *H.-parvus*-calibrated PTB, this normal interval is straddling the base of the Triassic [18,37,39], with the result that the lower half of the Griesbachian is overlapping with the Changxingian (e.g., [3,40]) and thus being Late Permian in age.

Whereas Tethyan magnetic records from the Triassic portion of the Griesbachian are indicated by the abovementioned thick normal magnetozone (e.g., [6,10]), the Boreal magnetostratigraphies additionally reveal a succeeding thick interval of predominantly reversed polarity (e.g., [5,8]). The resulting discrepancy concerning the size of the Griesbachian and the position of the Griesbachian–Dienerian boundary is most probably a consequence of the poorly defined substage boundaries of the corresponding Boreal sections, which above all is due to the relatively wide spacing of ammonite-rich horizons (e.g., [5,8]).

Nevertheless, the magnetic polarity patterns from the different Lower Triassic environments are in

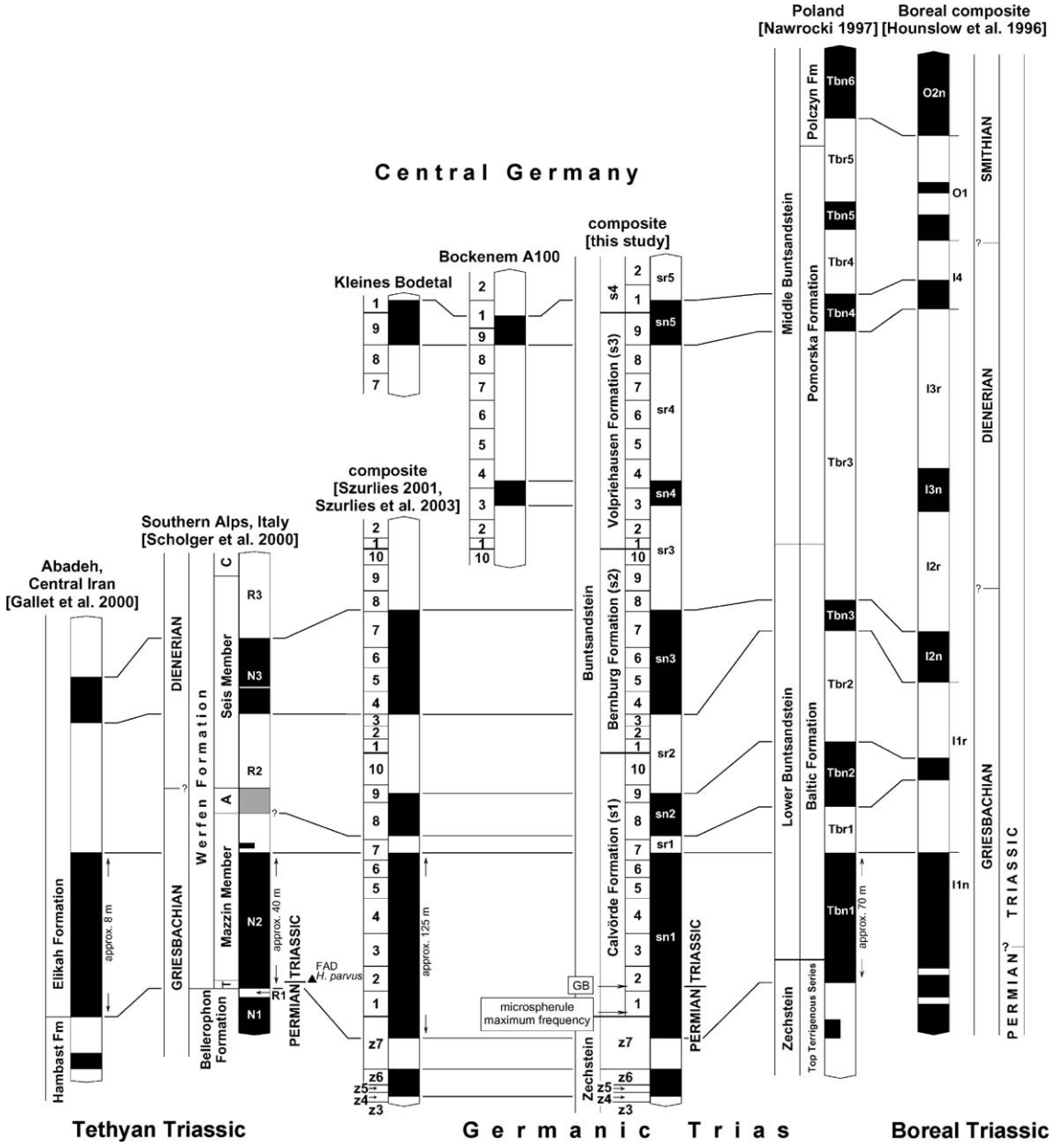


Fig. 8. Composite magnetic polarity of uppermost Zechstein to lowermost Detfurth Formation from Central Germany (in the middle) compiled from 11 sections and referred to the high-resolution cyclic stratigraphy [17,18]. Comparison with magnetostratigraphies from Poland [9] and the Boreal realm [8] (on the right) as well as from Central Iran [11] and Italy [10] (on the left). Position of the *H.-parvus*-calibrated PTB within the so-called "Graubankbereich" ("greengray interval"), GB. T=Tesero Oolite, A=Andraz Horizon, C=Campil Member. See Figs. 2 and 7 for key.

good accordance. The lowermost thick normal interval, which is visible in all Boreal magnetostratigraphies (e.g., I1n; [8]), correlates with magnetozone sn1, and the succeeding normal polarity zones sn2 to sn5 are most probably equivalents of the unnamed thin normal interval within I1r as well as I2n to I4n from the Boreal composite (Fig. 8).

The magnetic records from the Tethyan realm reveal a thick normal magnetozone as well as (e.g., N2; [10]) with the *H.-parvus*-calibrated PTB being located in its lowermost part, for example, in the lowermost Werfen Formation [10] and the lowermost Elikah Formation [11], respectively. Note that, in earlier studies, the base of the Boundary Clay was used as PTB (e.g., [6,11]).

As obtained from combined magnetostratigraphic–biostratigraphic correlation with the Tethyan realm, the base of the Dienerian is located in the uppermost Calvörde Formation (e.g., [3,18]), meaning that the base of the Dienerian, as used in the Tethys, is situated within the upper portion of magnetozone I1r [8] of the Boreal composite. In terms of magnetostratigraphy, the succeeding predominantly reversed Bernburg and Volpriehausen Formations may be equivalents of most of the Dienerian and the lowermost Smithian interval of the Tethyan and Boreal realms (e.g., [5,8]). It is succeeded by a remarkable thick normal polarity zone, which is spanning most of the Smithian in the Boreal records (e.g., O2n [8]). An ongoing study aims to precisely locate this normal interval within the high-resolution lithostratigraphy of the Buntsandstein. Presumably, this further first-order feature of the Scythian magnetostratigraphy may correlate with at least the Hardegsen Formation (e.g., [9,15]). Along with the thick normal polarity interval, which is straddling the base of the Triassic, both notably thick magnetozones form valuable anchors to distinctly faciliate an unambiguous correlation of the Buntsandstein high-resolution lithostratigraphy and the biostratigraphically calibrated marine stratotypes.

6. Conclusions

The ~150-m thick Volpriehausen Formation of Central Germany can be subdivided at least into nine small-scale fining-upward cycles, providing a robust high-resolution lithostratigraphic framework. Along with wireline logs (e.g., GR logs), they have been mapped successfully across much of the CEB. Moreover, the correlation is supported by prominent litho and gamma markers (e.g., Görtel sandstone). Alternatively, the cycles can be interpreted as more or less symmetrical base-level cycles, which are considered to reflect ~100-ka Milankovitch eccentricity cycles, providing more or less isochronous units, as suggested by the consistence of magnetozones and lithostratigraphic markers in the between-site comparison. From this assumption, it can be concluded that the duration of the Volpriehausen Formation is at least 0.9 Ma.

The paleomagnetic signal from the Volpriehausen Formation preserves a dual polarity magnetization, which is carried by hematite in the red-brown lithologies and by magnetite in the gray lithologies, respectively. It is allocated to two antipodal groups of directions (mean values Kleines Bodetal site: D=14.4°, I=24.7°, $\alpha 95$=3.8°). The preservation of a dual polarity magnetization close to the expected Early Triassic direction, the repetition of a comparable polarity pattern in both sections, the consistent correlation of lithostratigraphic units and magnetozones, and the lack of polarity changes in the place where magnetite became the magnetic carrier instead of hematite are the main arguments supporting an early acquisition of the characteristic magnetizations. In terms of magnetostratigraphy, two normal (sn4, sn5) and three reversed polarity zones (sr3 to sr5) are delineated. According to this assumption as well as from recently published data [17,18], the polarity intervals of the lowermost Triassic lasted ~0.1 to 0.7 Ma, with an average duration of ~0.3 Ma, which is comparable with the Late Cenozoic magnetozones (e.g., [41]).

In terms of magnetostratigraphy, the Lower Buntsandstein most probably corresponds to most of the Griesbachian and Dienerian, and the succeeding Volpriehausen Formation may correlate to the Late Dienerian to Early Smithian interval. For a more conclusive correlation and to remedy the discrepancies revealed for the upper portion of the Dienerian and Smithian, further paleomagnetic studies, in particular, in the Tethyan realm and the CEB are necessary.

Acknowledgements

The author thanks G.H. Bachmann, N.R. Nowaczyk, and H.-G. Röhling for valuable discussions. I am much indebted to the Wirtschaftsverband Erdöl- und Erdgasgewinnung (W.E.G.) for permission to publish the GR logs of Brüggen Z1 and Hohenassel Z1 wells. I thank Dennis V. Kent and an anony-

mous reviewer for constructive reviews of the manuscript. This study was funded by the Deutsche Forschungsgemeinschaft (project: SZ 104/1-1).

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